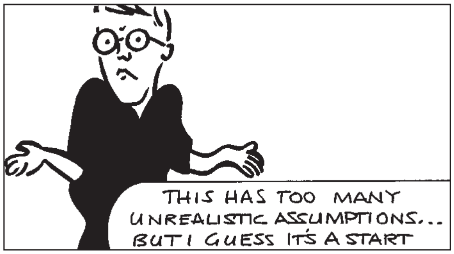
CHAPTER 24
RiskMetrics and CreditMetrics

The aim of this Chapter...

... is to describe JP Morgan's two popular methodologies for estimating risk associated with general movements of the financial markets, and with credit issues. We will be seeing how parameters are estimated and how the methodologies are implemented.

In this Chapter...

- the methodology of RiskMetrics for measuring value at risk
- the methodology of CreditMetrics for measuring a portfolio's exposure to default events



24.1 INTRODUCTION

In Chapter 22 I described the concept of the 'Value at Risk' (VaR) of a portfolio. I repeat the definition of VaR here: VaR is 'an estimate, with a given degree of confidence, of how much one can lose from one's portfolio over a given time horizon.' In that chapter I showed ways of calculating VaR (and some of the pitfalls of such calculations). Typically the data required for the calculations are parameters for the 'underlyings' and measures of a portfolio's current exposure to these underlyings. The parameters include volatilities and correlations of the assets, and, for longer time horizons, drift rates. The exposure of the portfolio is measured by the deltas, and, if necessary, the gammas (including cross derivatives) and the theta of the portfolio. The sensitivities of the portfolio are obviously best calculated by the owner of the portfolio, the bank. However, the asset parameters can be estimated by anyone with the right data. In October 1994 the American bank JP Morgan introduced the system **RiskMetrics** as a service for the estimation of VaR parameters. Some of this service is free, the datasets are available at `www.riskmetrics.com`, but the accompanying risk management software is not.

JP Morgan has also proposed a similar approach, together with a data service, for the estimation of risks associated with risk of default: **CreditMetrics**. CreditMetrics has several aims, two of these are the creation of a benchmark for measuring credit risk and the increase in market liquidity. If the former aim is successful then it will become possible to measure risks systematically across instruments and, at the very least, to make relative value judgments. From this follows the second aim. Once instruments, and in particular the risks associated with them, are better understood they will become less frightening to investors, promoting liquidity.

24.2 THE RISKMETRICS DATASETS

The RiskMetrics datasets are extremely broad and comprehensive. They are distributed over the internet. They consist of three types of data: one used for estimating risk over the time horizon of one day, the second having a one-month time horizon and the third has been designed to satisfy the requirements in the latest proposals from the Bank for International Settlements on the use of internal models to estimate market risk. The datasets contain estimates of volatilities and correlations for almost 400 instruments, covering foreign exchange, bonds, swaps, commodities and equity indices. Term-structure information is available for many currencies.

24.3 CALCULATING THE PARAMETERS THE RISKMETRICS WAY

A detailed technical description of the method for estimating financial parameters can be found at the JP Morgan web site. Here I only give a brief outline of major points.

24.3.1 Estimating volatility

The volatility of an asset is measured as the annualized standard deviation of returns. There are many ways of taking this measurement. The simplest is to take data going back a set period, three months, say, calculate the return for each day (or over the typical timescale at which you will be rehedging) and calculate the sample standard deviation of this data. This will result in a time series of three-month volatility.[1] This approach gives equal weight to all of the observations over the previous three months. This estimate of volatility on day i is calculated as

$$\sigma_i = \sqrt{\frac{1}{\delta t(M-1)} \sum_{j=i-M+1}^{i} (R_j - \overline{R})^2},$$

where δt is the time step (typically one day), M is the number of days in the estimate (approximately 63 in three months), R_j is the return on day j and $\overline{R}$ is the average return over the previous M days. If δt is small then we can in practice neglect $\overline{R}$.

This measurement of volatility has two major drawbacks. First, it is not clear how many days' data we should use; what happened three months ago may not be relevant to today. But the more data we have the smaller will be the sampling error if the volatility really has not changed in that period. Second, a large positive or negative return on one day will be felt in this historical volatility for the next three months. At the end of this period the volatility will apparently drop suddenly, yet there will have been no underlying change in market conditions; the drop will be completely spurious. Thus, the volatility measured in this way will show 'plateauing.' A typical 30-day volatility plot, with plateauing, is shown in Figure 24.1.

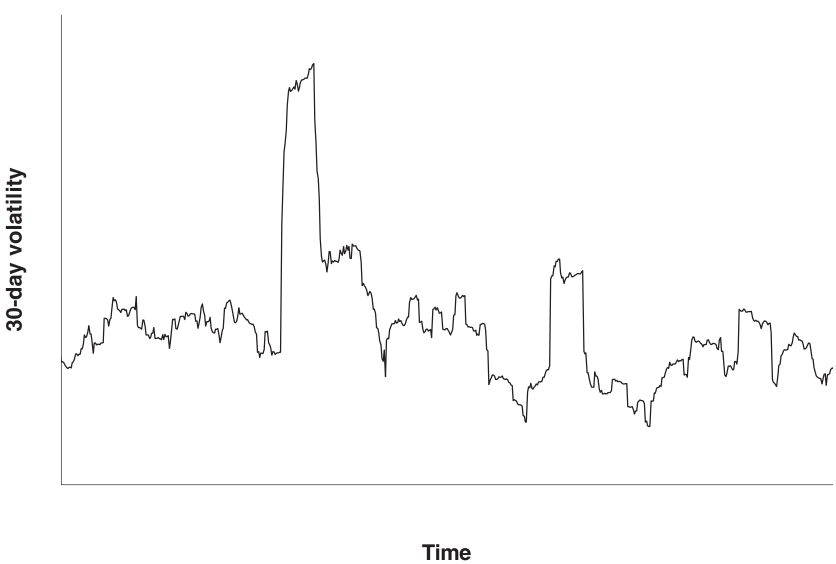


Figure 24.1 Thirty-day volatility.

[1] Which may or may not bear any resemblance to three-month implied volatility.

In RiskMetrics the volatility is measured as the square root of a variance that is an exponential moving average of the square of price returns. This ensures that any individual return has a gradually decreasing effect on the estimated volatility, and plateauing does not occur. This volatility is estimated according to

$$\sigma_i = \sqrt{\frac{1-\lambda}{\delta t} \sum_{j=-\infty}^{i} \lambda^{i-j} R_j^2},$$

where λ represents the weighting attached to the past volatility versus the present return (and we have neglected the mean of R, assuming that the time horizon is sufficiently small). This difference in weighting is more easily seen if we write the above as

$$\sigma_i^2 = \lambda \sigma_{i-1}^2 + (1-\lambda)\frac{R_i^2}{\delta t}.$$

The parameter λ has been chosen by JP Morgan as either 0.94 for an horizon of one day and 0.97 for an horizon of one month. Another possibility is to choose λ to minimize the difference between the squares of the historical volatility and an implied volatility.

The spreadsheet in Figure 24.2 shows how to calculate such an exponentially weighted volatility. As can be seen from the plot, it is much 'better behaved' than the uniformly weighted version.

Exponentially weighted moving averages

24.3.2 Correlation

The estimation of correlation is similar to that of volatility. To calculate the covariance σ_{12} between assets 1 and 2 we can take an equal weighting of returns from the two assets over the last M days:

$$\sigma_{12_i} = \sqrt{\frac{1}{\delta t(M-1)} \sum_{j=i-M+1}^{i} (R_{1_j} - \overline{R}_1)(R_{2_j} - \overline{R}_2)}.$$

Again, this measure shows spurious sudden rises and falls because of the equal weighting of all the returns.

Alternatively, we can use an exponentially weighted estimate:

$$\sigma_{12_i}^2 = \lambda \sigma_{12_{i-1}}^2 + (1-\lambda)\frac{R_{1_i} R_{2_i}}{\delta t}.$$

There are problems with the estimation of covariance due to the synchronicity of asset movements and measurement. Two assets may be perfectly correlated but because of their measurement at different times they may appear to be completely uncorrelated. This is a problem when using data from markets in different time zones. Moreover, there is no guarantee that the exponentially weighted covariances give a positive-definite matrix.

24.4 THE CREDITMETRICS DATASET

The CreditMetrics dataset is available free of charge from `www.jpmorgan.com`. The CreditMetrics methodology is also described in great detail at that site. The dataset consists of four data types: yield curves, spreads, transition matrices and correlations. Before reading the following sections the reader should be comfortable with the concept of credit rating, see Chapter 23.

	A	B	C	D	E	F	G	H
1	**Start volatility**	0.3		**Date**	**Stock**	**Returns**	**Vol^2**	**Volatility**
2	**Lambda**	0.97		1-Jan-85	218.32		0.09	0.3
3				2-Jan-85	217.16	-0.005313	0.087513	0.295827
4		= (E3–E2)/E2		3-Jan-85	215.24	–0.008841	0.085479	0.292368
5				4-Jan-85	215.24	0.000000	0.082915	0.287949
6		= B1*B1		7-Jan-85	217.16	0.008920	0.081029	0.284655
7				8-Jan-85	220.25	0.014229	0.080129	0.28307
8	= B2*G7 + (1-B2)*F8*F8*252			Jan-85	224.87	0.020976	0.081051	0.284695
9				10-Jan-85	224.87	0.000000	0.07862	0.280392
10				11-Jan-85	224.1	–0.003424	0.07635	0.276314
11		= SQRT(G11)		14-Jan-85	219.09	–0.022356	0.077838	0.278994
12				15-Jan-86	219.09	0.000000	0.075502	0.274777
13				16-Jan-85	222.17	0.014058	0.074731	0.273371
14				17-Jan-85	226.02	0.017329	0.07476	0.273422
15				18-Jan-85	226.02	0.000000	0.072517	0.26929
16				21-Jan-85	226.79	0.003407	0.070429	0.265385
17				22-Jan-85	234.49	0.033952	0.077031	0.277545
18					57	–0.008188	0.075227	0.274275
19					65	0.013243	0.074296	0.272573
20					27	0.019605	0.074973	0.273812
21					.5	–0.003205	0.072802	0.269818
22					72	–0.024134	0.075021	0.273899
23					35	0.041203	0.085605	0.292583
24					12	0.003164	0.083112	0.288292
25					04	0.007865	0.081086	0.284757
26					04	0.000000	0.078654	0.280453
27					39	0.015648	0.078145	0.279545
28					97	–0.007683	0.076247	0.276129
29				7-Feb-85	247.97	0.000000	0.07396	0.271956
30				8-Feb-85	249.12	0.004638	0.071904	0.268149
31				11-Feb-85	243.35	–0.023162	0.073802	0.271666
32				12-Feb-85	238.34	–0.020588	0.074792	0.273482
33				13-Feb-85	239.5	0.004867	0.072728	0.269681
34				14-Feb-85	238.34	–0.004843	0.070723	0.265938
35				15-Feb-85	235.65	–0.011286	0.069565	0.263751
36				18-Feb-85	232.57	–0.013070	0.068769	0.262239
37				19-Feb-85	234.49	0.008256	0.067221	0.259271
38				20-Feb-85	235.65	0.004947	0.06539	0.255714
39				21-Feb-85	236.42	0.003268	0.063509	0.252009
40				22-Feb-85	235.65	–0.003257	0.061684	0.248362
41				25-Feb-85	234.49	–0.004923	0.060016	0.244982
42				26-Feb-85	235.65	0.004947	0.058401	0.241663
43				27/02/85	232.57	–0.013070	0.05794	0.240708

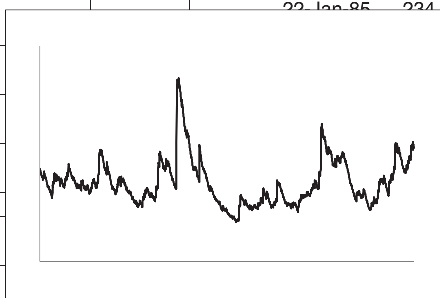


Figure 24.2 Spreadsheet to calculate an exponentially weighted volatility.

24.4.1 Yield curves

The CreditMetrics yield curve dataset consists of the *risk-free* yield to maturity for major currencies. In Figure 24.3 is shown an example of these risk-free yields. The dataset contains yields for maturities of one, two, three, five, seven, 10 and 30 years. For example, from the yield curve dataset we have information such as the yield to maturity for a three-year US dollar bond is 6.12%.

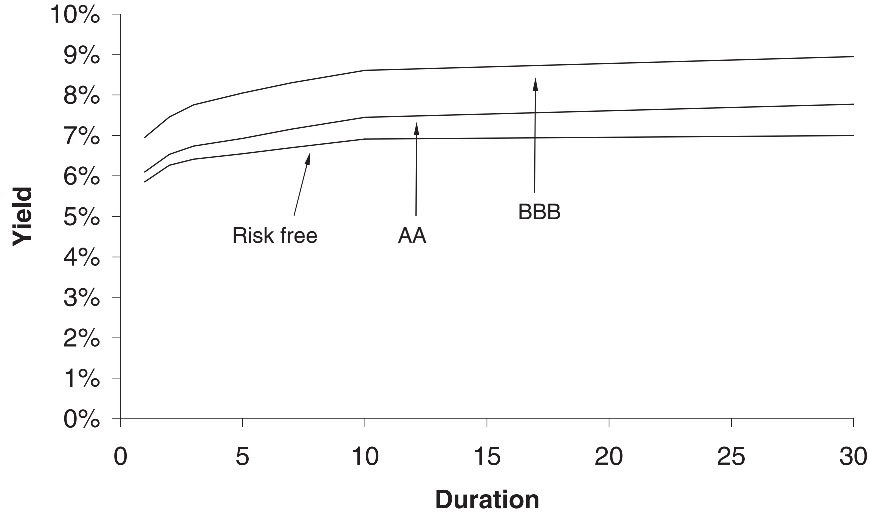


Figure 24.3 Risk-free and two risky yield curves.

24.4.2 Spreads

For each credit rating, the dataset gives the spread above the riskless yield for each maturity. In Figure 24.3 is shown a typical riskless US yield curve, and the yield on AA and BBB bonds. For example, we may be given that the spread for an AA bond is 0.54% above the riskless yield for a three-year bond. Observe that the riskier the bond the higher the yield; the yield on the BBB bond is everywhere higher than that on the AA bond which is in turn higher than the risk-free yield. This higher yield for risky bonds is compensation for the possibility of not receiving future coupons or the principal.

24.4.3 Transition matrices

The concept of the transition matrix has been discussed in Chapter 23. In the CreditMetrics framework, the transition matrix has as its entries the probability of a change of credit rating at the end of a given time horizon, for example the probability of a upgrade from AA to AAA might be 5.5%. The time horizon for the CreditMetrics dataset is one year. Unless the time horizon is very long, the largest probability is typically for the bond to remain at its initial rating, let's say that the probability of staying at AA is 87% in this example. We discussed transition matrices in depth in Chapter 23.

24.4.4 Correlations

In the risk-free yield, the spreads and the transition matrix, there is sufficient information for the CreditMetrics method to derive distributions for the possible future values of a single bond. I show how this is done in the next section. However, when we come to examine the behavior of a portfolio of risky bonds, we must consider whether there is any relationship between the rerating or default of one bond and the rerating or default of another. In other words, are bonds issued by different companies or governments in some sense correlated? This is where the CreditMetrics correlation dataset comes in. This dataset gives the correlations between major indices in many countries.

Each company issuing bonds has the return on its stock decomposed into parts correlated with these indices and a part which is specific to the company. By relating all bond issuers to these indices we can determine correlations between the companies in our portfolio. We will see how this is used in practice later in this chapter.

24.5 **THE CREDITMETRICS METHODOLOGY**

The CreditMetrics methodology is about calculating the possible values of a risky portfolio at some time in the future (the time horizon) and estimating the probability of such values occurring. Let us consider just a single risky bond currently rated AA. Suppose that the bond is zero coupon, with a maturity of three years and we want to know the possible state of this investment in one year's time. The yield to maturity on this instrument might be 6.12%, for a three-year riskless bond, plus 0.54% for the spread for a three-year AA-rated bond. The total yield is therefore 6.66%, giving a price of 0.819.

The value of the bond will fluctuate between now and one year's time for each of three reasons: the passage of time, the evolution of interest rates and the possible regrading of the bond. Let us take these three points in turn.

First, because of the passage of time our three-year bond will be a two-year bond in one year. But what will be the yield on a two-year bond in one year's time? This is the second point. The assumption that is made in CreditMetrics is that the forward rates do not change between today and the time horizon ('rolling down the curve'). From the yields that we have today we can calculate the forward rates that apply between now and one year, between one and two years, between two and three years, etc. This calculation is described in Chapter 14. We can calculate the value of the bond after one year, suppose it is 0.882. But why should the bond still be rated AA at that time? This is the third point. From our transition matrix we see that the probability of the bond's rating staying AA is 87%. So, there is an 87% chance that the bond's value will be 0.882. We can similarly work out the value of the bond in one year if it is rated AAA, A, BBB, etc. using the relevant forward rates and spreads that we assume will apply in one year's time. And each of these has a probability of occurring that is given in the transition matrix. A probability distribution of the possible bond values is shown in Figure 24.4.

This, highly skewed, distribution tells us all we need to know to determine the risk in this particular bond. We can, for example, calculate the expected value of the bond.

24.6 **A PORTFOLIO OF RISKY BONDS**

We have seen how to apply the CreditMetrics methodology to a single risky bond, to apply the ideas to a portfolio of risky bonds is significantly harder since it requires the knowledge of any relationship between the different bonds. This is most easily measured by some sort of correlation.

Suppose that we have a portfolio of two bonds. One, issued by ABC, is currently rated AA and the other, issued by XYZ, is BBB. We can calculate, using the method above, the value of each of these bonds at our time horizon for each of the possible states of

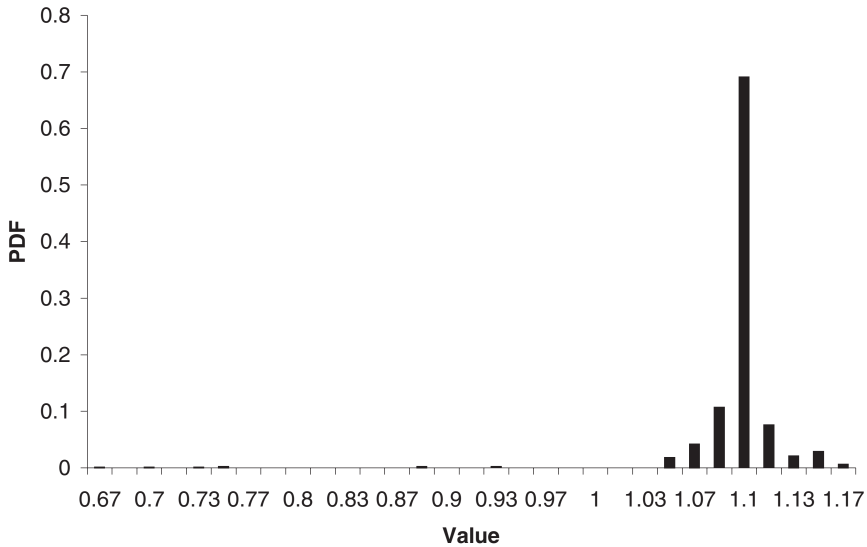


Figure 24.4 The probability distribution for the bond's value after one year.

the two bonds. If we assume that each bond can be in one of eight states (AAA, AA, ..., CCC, Default) there are $8^2 = 64$ possible joint states at the time horizon. To calculate the expected value of our portfolio and standard deviation we need to know the probability of each of these joint states occurring. This is where the correlation comes in.

There are two stages to determining the probability of any particular future joint state:

1. Calculate the correlations between bonds.
2. Calculate the probability of any joint state.

Stage 1 is accomplished by decomposing the return on the stock of each issuing company into parts correlated with the major indices.

24.7 CREDITMETRICS MODEL OUTPUTS

CreditMetrics is, above all, a way of measuring risk associated with default issues. From the CreditMetrics methodology one can calculate the risk, measured by standard deviation, of the risky portfolio over the required time horizon. Because of the risk of default the distribution of returns from a portfolio exposed to credit risk is highly skewed, as in Figure 24.4. The distribution is far from being Normal. Thus ideas from simple portfolio theory must be used with care. Although it may not be a good absolute measure of risk in the classical sense, the standard deviation is a good indicator of relative risk between instruments or portfolios.

24.8 SUMMARY

This chapter has outlined some of the methodologies for competing and complementary Value at Risk measures. With something as important as Value at Risk there is an obvious case to be made for exploring all of the possible VaR measures to build up as accurate a profile as possible of the dangers lurking in your portfolio. In the next chapter we look at measuring and reducing the risk in stock market crashes.

FURTHER READING

- Download the datasets and very detailed descriptions of the RiskMetrics and CreditMetrics methodologies from `www.riskmetrics.com`.
- Alexander (1996a) is a critique of RiskMetrics as a risk measurement tool.
- Shore (1997) describes and implements the CreditMetrics methodology.